Qingcheng Zeng

Ph.D. Candidate · LLM Agents, Reinforcement Learning & Information Retrieval

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Profile

Ph.D. candidate in Computational Linguistics with industry experience in agentic reinforcement learning, ML infrastructure, and production retrieval at Alibaba International, Snowflake, and Tencent AI Lab. Research spans instruction-following retrieval, embeddings, trustworthy LLMs, multilingual NLP, and benchmark design, with publications at ACL, EMNLP, NeurIPS, EACL, CIKM, and IJCAI.

Technical Expertise

Applied ML: Python, PyTorch, Hugging Face Transformers; LLM post-training and experimental analysis.
Agentic RL: GRPO, PPO, SAO, process reward modeling; slime and Miles frameworks.
Search & Evaluation: information retrieval, dense embeddings, instruction following, multilingual retrieval, calibration, robustness, and benchmark design.

Industry Experience

Applied Scientist Intern — Alibaba International, Accio Agent Team June 2026 – Sep 2026
ML Infrastructure & Agentic Reinforcement Learning

- Contributed ML infrastructure and RL algorithm development to Occamy-1.0, an open-weight 35B-A3B co-work model; trained long-horizon tool-use agents with Single-Rollout Asynchronous Optimization (SAO).
- The released model improved Claw-Eval Pass^3 from 54.8 to 71.4 over its base checkpoint while using 19.5% fewer tokens under the same evaluation protocol (technical report).

Research Scientist Intern — Snowflake, Cortex Search AI Team March 2026 – June 2026
Remote

- Built a five-dataset SQL schema-retrieval benchmark and corpus-adaptive embeddings, raising average Recall@10 from 60.4 to 75.6 (paper).

Research Scientist Intern — Snowflake, Cortex Search AI Team Sep 2025 – Dec 2025
Menlo Park, CA

- Shipped instruction-following retrieval for Cortex Search and led Dual-View Training (*EMNLP 2026 Main*), improving FollowIR p-MRR by 57.5% on both tested backbones.

Research Scientist Intern — Tencent AI Lab America June 2025 – Aug 2025
Bellevue, WA

- Developed verified Critical Step Optimization for LLM agents, improving GAIA-Text-103 accuracy by 37% over SFT while supervising only 16% of trajectory steps (*Findings of ACL 2026*).

Selected Research

- **Retrieval alignment & embedding quality:** built instruction-following retrievers with dual-view training and diagnosed spread-out regularization failures in dense embeddings (*EMNLP 2026 Main*).
- **Reliable agents & RAG:** analyzed and mitigated miscalibration in tool-use agents (*ACL 2026 Oral*) and studied noise-aware confidence calibration for RAG (*Findings of EMNLP 2026*).
- **Multilingual & code-switching search:** created cross-lingual benchmarks to expose failure modes in retrievers and deep-research agents (*Findings of ACL 2026*; arXiv 2026).
- **Model behavior & risk analysis:** studied linguistic bias across encoder- and decoder-only models (*EMNLP 2026 Main*) and confidence reliability in reasoning models (*EMNLP 2025*).

Education & Recognition

Northwestern University — Ph.D. in Computational Linguistics, expected 2026 Advisors: Rob Voigt, Kaize Ding
University of Manchester — B.A. in Linguistics, 2021 • **Zhejiang University** — B.A. in Linguistics, 2021
Recognition: Gold Reviewer, ICML 2026; Excellent Reviewer, KDD 2025 Applied Data Science Track. **Languages:** Mandarin Chinese (native); English (professional proficiency).